LAB SHEET: WEB DESIGN TECHNOLOGY (LAB 3.2)

TITLE: CONSTRUCTING AN INTERACTIVE STUDENT PORTFOLIO WITH BOOTSTRAP 5

TOPIC: 3.2 Construct Interactive Web Elements with Bootstrap

STUDENT INFORMATION

Name	Registration No	Date
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1. LEARNING OUTCOMES

By the end of this session, you will be able to:

- **Implement** professional Bootstrap Modals for data entry (3.2.1).
- **Enhance** User Experience (UX) using interactive Tooltips and Popovers (3.2.2).
- **Construct** an Advanced Carousel featuring a custom progress bar and thumbnail navigation (3.2.3).

2. THEORETICAL OVERVIEW

To create a high-fidelity interface, we will combine Bootstrap 5 with Custom JavaScript:

- **The Modal:** A "child" window that pops up to focus user attention.
- **Popper.js Components:** Tooltips (hover) and Popovers (click). *Note: These are not auto-enabled; we must "wake them up" using JavaScript!*
- **Carousel Events:** We use the `slid.bs.carousel` event to sync our progress bar and thumbnails in real-time.

3. STEP-BY-STEP IMPLEMENTATION

Step 1: Document Setup & Design Language

Create a file named `portfolio.html`. We'll define a "Primary Purple" theme to make your portfolio look modern and premium.

Action: Setup the `<head>` and define the CSS variables:

HTML

```
<style>
    :root {
        --primary-purple: #5d46b8;
        --bg-soft: #f0f4f8;
    }
    body { background-color: var(--bg-soft); font-family: 'Segoe UI',
sans-serif; }
    .navbar-custom { background-color: var(--primary-purple); }
    .card { border: none; border-radius: 12px; box-shadow: 0 4px 10px
rgba(0, 0, 0, 0.05); margin-bottom: 20px; }

    /* Advanced Carousel Styling */
    .carousel-demo img { height: 400px; object-fit: cover; border-
radius: 8px; }
    .active-thumb { border: 3px solid var(--primary-purple)
!important; opacity: 1 !important; }
    .thumbs img { width: 100px; height: 60px; object-fit: cover;
cursor: pointer; opacity: 0.6; transition: 0.3s; }

    #btnTop { display: none; position: fixed; bottom: 30px; right:
30px; z-index: 99;
              background-color: var(--primary-purple); color: white;
border: none;
              padding: 15px; border-radius: 50%; transition: 0.3s; }
</style>
```

📸 **Checkpoint 1:** Screenshot your blank page/setup and explain why we use CSS `:root` variables.

Step 2: Profile Section (Tooltips & Popovers)

Let's add your identity! We will use **Tooltips** for skill badges and a **Popover** for your bio.

Action: Insert the Navbar and Profile Card into the `<body>`:
HTML

```
<nav class="navbar navbar-expand-lg navbar-dark navbar-custom mb-4">
    <div class="container justify-content-center">
        <ul class="navbar-nav">
            <li class="nav-item"><a class="nav-link active"
href="#"><i class="fas fa-user-circle"></i> My Profile</a></li>
            <li class="nav-item"><a class="nav-link" href="#"><i
class="fas fa-laptop-code"></i> Web Tech</a></li>
        </ul>
    </div>
</nav>

<div class="container pb-5">
    <div class="row">
        <div class="col-lg-4">
            <div class="card p-3 text-center">
                <img src="https://via.placeholder.com/120"
class="rounded-3 mb-3 shadow-sm mx-auto" alt="Profile">
                <h4 class="fw-bold mb-1">Your Name</h4>
                <p class="text-muted small">ID: 10DITXXXXXX</p>
```

```
                <div class="d-flex flex-wrap justify-content-center
gap-2 mt-2">
                    <span class="badge bg-primary" data-bs-
toggle="tooltip" title="Final Year Student">Malaysia</span>
                    <span class="badge bg-secondary" data-bs-
toggle="tooltip" title="HTML/CSS/JS">Frontend Dev</span>
                </div>
            </div>
            <div class="card p-3">
                <h5><i class="fas fa-info-circle text-primary me-
2"></i>Quick Bio</h5>
                <button type="button" class="btn btn-sm btn-outline-
primary" data-bs-toggle="popover"
                        data-bs-trigger="focus" title="Academic
Background"
                        data-bs-content="Currently pursuing Diploma
in IT.">View Detailed Bio</button>
            </div>
        </div>
```

📸 **Checkpoint 2:** Show your Profile Section. Hover your mouse over the "Malaysia" badge to show the Tooltip.

Step 3: The Advanced Project Carousel

Time to show off your work! This carousel includes a **Progress Bar** and **Thumbnails**.

Action: Add this inside the `col-lg-8` div:
HTML

```
            <div class="col-lg-8">
                <div class="card shadow-sm border-0">
                    <div class="card-body p-4">
                        <header class="mb-4 text-center">
                            <h2 class="h4 fw-bold text-
primary">Project Showcase</h2>
                            <p class="text-muted small">Hover to
pause, click thumbnails to navigate.</p>
                        </header>

                        <div id="demoCarousel" class="carousel slide
carousel-demo" data-bs-ride="carousel">
                            <div class="carousel-inner shadow-sm
rounded">
                                <div class="carousel-item active">
                                    <img
src="https://images.unsplash.com/photo-1551434678-
e076c223a692?auto=format&fit=crop&w=1200&h=600"
                                        class="d-block w-100"
alt="Database">
                                    <div class="carousel-caption d-
none d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>User Interface
Design</h5>
                                        <p>Crafting modern layouts
with Bootstrap 5 Grid and Utility classes.</p>
```

```
                                    </div>
                                </div>
```

```
<div class="carousel-item">
                                    <img
src="https://images.unsplash.com/photo-1498050108023-
c5249f4df085?auto=format&fit=crop&w=1200&h=600"
                                        class="d-block w-100" alt="JS
Interactivity">
                                    <div class="carousel-caption d-none
d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>Database Integration</h5>
                                        <p>Dynamic data management with
PHP & MySQL structural logic.</p>
                                    </div>
                                </div>
                                <div class="carousel-item">
                                    <img
src="https://images.unsplash.com/photo-1507721999472-
8ed4421c4af2?q=80&w=1200&h=600&auto=format&fit=crop"
                                        class="d-block w-100" alt="JS
Interactivity">
                                    <div class="carousel-caption d-none
d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>JavaScript Interactivity</h5>
                                        <p>Mastering DOM manipulation and
Event Listeners for seamless UX.</p>
                                    </div>
                                </div>
                            </div>
```

```
                            <button class="carousel-control-prev"
type="button" data-bs-target="#demoCarousel"
                                data-bs-slide="prev">
                                <span class="carousel-control-prev-
icon"></span>
                            </button>
                            <button class="carousel-control-next"
type="button" data-bs-target="#demoCarousel"
                                data-bs-slide="next">
                                <span class="carousel-control-next-
icon"></span>
                            </button>
                        </div>

                        <div class="progress mt-3" style="height:6px;">
                            <div id="carouselProgress"
                                class="progress-bar progress-bar-striped
progress-bar-animated bg-info"
                                style="width:0%"></div>
                        </div>

                        <div class="thumbs d-flex justify-content-center
gap-2 mt-4">
                            <img src="https://images.unsplash.com/photo-
1551434678-e076c223a692?auto=format&fit=crop&w=1200&h=600" class="img-
thumbnail active-thumb"
                                data-slide-to="0">
                            <img src="https://images.unsplash.com/photo-
1498050108023-c5249f4df085?auto=format&fit=crop&w=1200&h=600" class="img-
thumbnail" data-slide-to="1">
```

```
                            <img src="https://images.unsplash.com/photo-
1507721999472-8ed4421c4af2?q=80&w=1200&h=600&auto=format&fit=crop"
class="img-thumbnail" data-slide-to="2">
                        </div>
                    </div>
                </div>
```

```
<div class="card p-4 border-start border-info border-5 shadow-sm">
                    <div class="d-flex justify-content-between align-
items-center">
                        <div>
                            <h5 class="fw-bold mb-1">Class
Registration</h5>
                            <p class="text-muted small mb-0">Join the Web
Design Technology session for Semester 2.</p>
                        </div>
                        <button type="button" class="btn btn-info text-
white fw-bold px-4" data-bs-toggle="modal"
                            data-bs-target="#registrationModal">
                            <i class="fas fa-user-plus me-2"></i>Register
                        </button>
                    </div>
                </div>
            </div>
        </div>
    </div>
```

Step 4: Registration Modal & Footer

We will add a "Call to Action" button that opens a beautiful modal.

Action: Add the Modal at the very bottom of your `<body>`:
HTML

```
<div class="modal fade" id="registrationModal" tabindex="-1">
    <div class="modal-dialog modal-dialog-centered">
        <div class="modal-content border-0">
            <div class="modal-header bg-info text-white">
                <h5 class="modal-title">Workshop Enrollment</h5>
                <button type="button" class="btn-close btn-close-
white" data-bs-dismiss="modal"></button>
            </div>
            <form id="regForm">
                <div class="modal-body p-4">
                    <label class="form-label fw-bold">Full
Name</label>
                    <input type="text" class="form-control mb-3"
id="formName" required>
                    <label class="form-label fw-bold">Student
ID</label>
                    <input type="text" class="form-control"
placeholder="DITXXXX" required>
                </div>
                <div class="modal-footer">
                    <button type="submit" class="btn btn-info text-
white px-4">Submit Application</button>
                </div>
```

```
            </form>
        </div>
    </div>
</div>

    <button onclick="topFunction()" id="btnTop" title="Go to
top"><i class="fas fa-arrow-up"></i></button>

    <footer class="navbar-custom py-4 mt-5">
        <div class="container text-center text-white">
                <p class="mb-1">© Copyright 2026 - JTMK
Polytechnic</p>
            <p class="mb-0" style="font-size: 0.85rem; opacity:
0.8;">Prepared by: Shu Haila Mohd Yusof (Reg. ID.:
10DIT26F1705)</p>
        </div>
    </footer>
```

Step 5: Bringing It to Life (JavaScript)

This is where the magic happens! We will initialize all components.

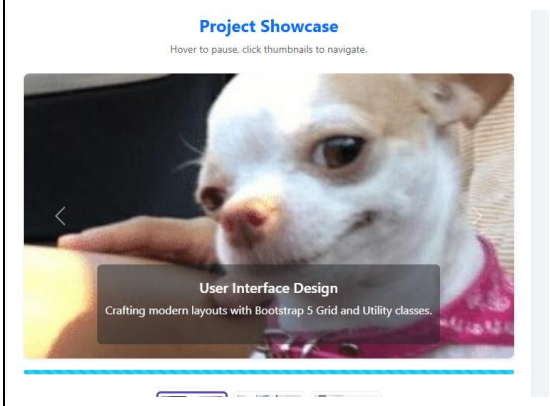
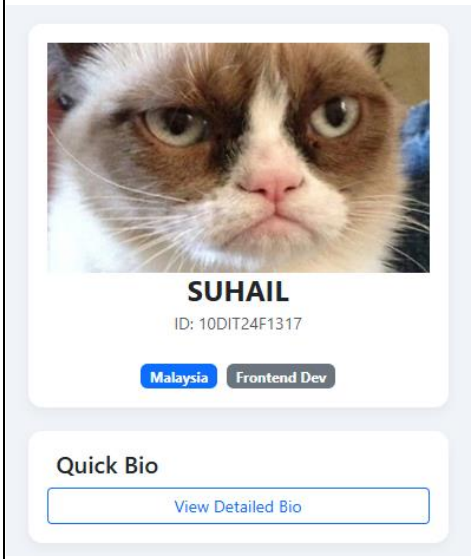
Action: Add the script before the `</body>` tag:
JavaScript

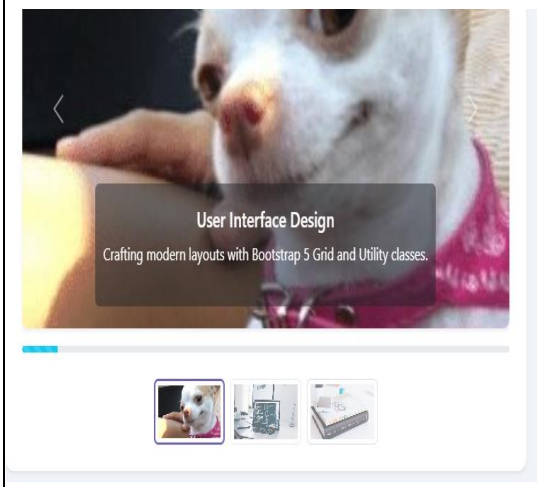
```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></s

    <script>
        (function () {
            // Initialize Tooltips & Popovers (3.2.2)
            const tooltipTriggerList = document.querySelectorAll('[data-bs-toggle="too
            tooltipTriggerList.forEach(el => new bootstrap.Tooltip(el));

            const popoverTriggerList = document.querySelectorAll('[data-bs-toggle="pop
            popoverTriggerList.forEach(el => new bootstrap.Popover(el));

            // Advanced Carousel Logic (3.2.3)
            const carouselEl = document.querySelector('#demoCarousel');
            const intervalMs = 4000;
            const bar = document.getElementById('carouselProgress');
            const thumbs = document.querySelectorAll('.thumbs img');
             const carousel = new bootstrap.Carousel(carouselEl, { interval: interval
'hover' });

            let progressTimer;
            function startProgress() {
                clearInterval(progressTimer);
                bar.style.width = '0%';
                const start = Date.now();
                progressTimer = setInterval(() => {
                    const elapsed = Date.now() - start;
                    const pct = Math.min(100, (elapsed / intervalMs) * 100);
                    bar.style.width = pct + '%';
                    if (pct >= 100) clearInterval(progressTimer);
```

```
                }, 50);
            }

            // Sync Thumbnails and Progress Bar
            carouselEl.addEventListener('slid.bs.carousel', (e) => {
                thumbs.forEach(t => t.classList.remove('active-thumb'));
                thumbs[e.to].classList.add('active-thumb');
                startProgress();
            });

            // Thumbnail Navigation
            thumbs.forEach((thumb, idx) => {
                thumb.addEventListener('click', () => carousel.to(idx));
            });

            carouselEl.addEventListener('slide.bs.carousel', () => clearInterval(progre

            startProgress(); // Start progress for the first slide

            // Back to Top logic
            let mybutton = document.getElementById("btnTop");
            window.onscroll = () => { mybutton.style.display = (window.scrollY > 100) ?
"none"; };

            // Registration Feedback
            document.getElementById('regForm').addEventListener('submit', function (e)
                e.preventDefault();
                 alert("Success! Registration for " + document.getElementById('formName
" is complete.");
                bootstrap.Modal.getInstance(document.getElementById('registrationModal
            });
        })();

        function topFunction() {
            window.scrollTo({ top: 0, behavior: 'smooth' });
        }
    </script>
```

4. STEP-BY-STEP VALIDATION:

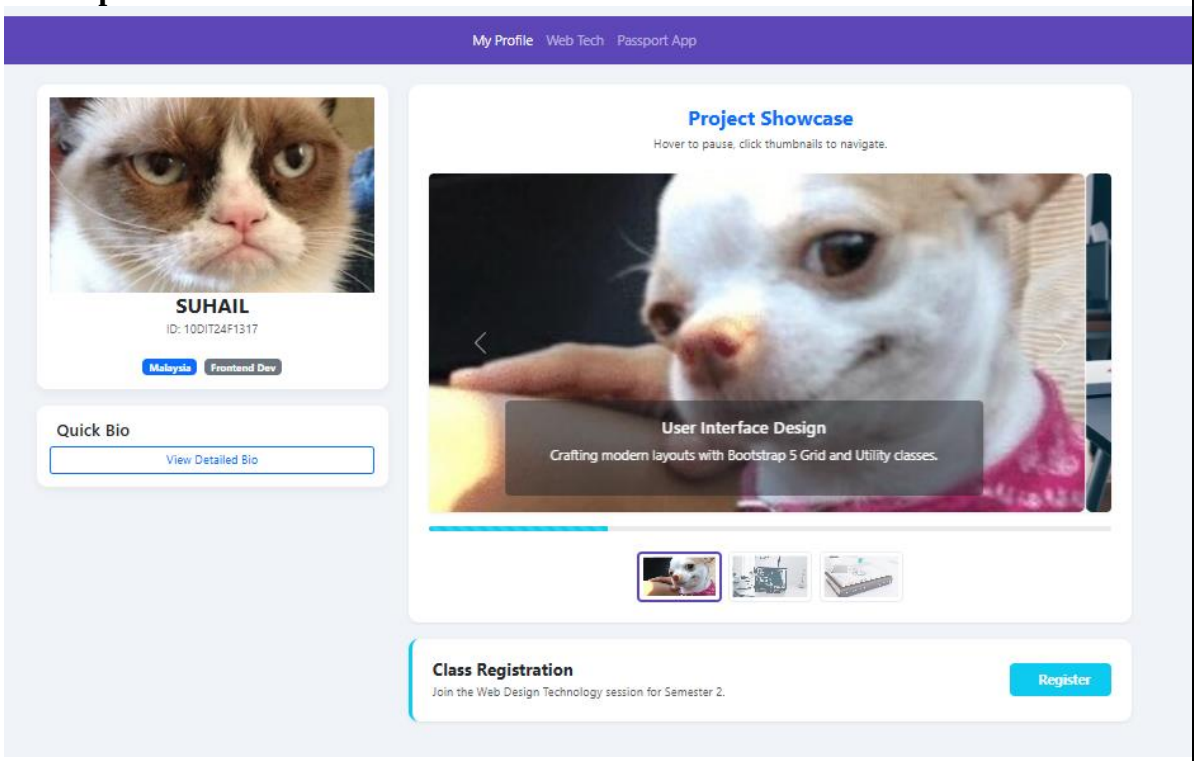
Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1: Setup		The `.card` class is used to design content cards. It removes the border, adds rounded corners, applies a soft shadow effect, and creates spacing between cards so that the layout looks clean and modern. The `.carousel-demo img` style controls the images in the carousel. It sets a fixed height for the images, keeps their proportions using `object-fit`, and adds rounded corners for a smoother appearance. The `.active-thumb` class highlights the currently selected thumbnail image by adding a purple border and making the image fully visible. The `.thumbs img` style is used for small thumbnail images. It sets their size, makes them clickable, slightly reduces their opacity, and adds a smooth transition effect when users interact with them.
Step 2: Profile		**Tooltips** display a small message when the user **moves the mouse over an element**. In the code, the tooltip is added using `data-bs-toggle="tooltip"`. The text shown in the tooltip comes from the `title` attribute. **Popovers** show **more detailed information** than tooltips. They usually appear when the user **clicks a button**. In the code, the popover is added using `data-bs-toggle="popover"`. The `title` shows the popover heading, and `data-bs-content` shows the message inside the popover.

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 3: The Advanced Project Carousel		The section `col-lg-8` is used to display the **Project Showcase**. Inside it, a **card** is used to organize the content neatly. At the top, there is a title **"Project Showcase"** and a short instruction for users. Next, a **carousel (slideshow)** is created. The carousel shows different images about projects such as **User Interface Design, Database Integration, and JavaScript Interactivity**. Each image has a caption that explains the project topic. Users can move between slides using the **previous and next buttons**. Below the carousel, a **progress bar** shows the progress of the slide changing. After that, **thumbnail images** are provided so users can click a small image to go directly to that slide.

5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View



Mobile View

6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: `RegNo_Date.zip`.
2. **Report:** Create a PDF containing all screenshots and explanations: `Report_RegNo_Date.pdf`.

Happy Coding! Success is in your hands! 🌟